3D Computer Vision

CS 532 study textbook

Oct 22, 2026

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From images to geometry

How can photographs reveal the shape of a scene and the motion of a camera? This book develops the geometric ideas behind that question through the first six CS 532 lectures.

Read in lecture order, beginning with the *notation guide* and *prerequisite refresher*. Worked examples will connect the lecture slides to the course quizzes. The *quiz index* shows the planned placements.

> **Note**
>
> This edition establishes the book's structure and authoring conventions. Lecture chapters are planning outlines; their explanations and examples are not yet written. Completed material will be added one lecture at a time.

How to study a section

1. Start with the visual question and explain the intuition in your own words.
2. Read the formula's assumptions and symbol definitions before following the derivation.
3. Work through the application procedure, then attempt the example without its solution.
4. Use the chapter checklist and formula reference to identify what still needs practice.

The website and printable edition are built from the same Markdown pages. Essential explanations and solutions belong in both editions.

Before you begin

NOTATION AND COORDINATE CONVENTIONS

These conventions apply throughout the book. A source using another convention will be translated explicitly.

Symbol	Meaning	Shape / units
$\mathbf{X}$	Homogeneous 3D point	4×1; scale arbitrary
$\mathbf{x}$	Homogeneous image point	3×1; scale arbitrary
(u, v)	Inhomogeneous pixel coordinates	pixels
$\mathbf{X}_c$	Euclidean camera point $(X_c, Y_c, Z_c)^T$	3×1; one consistent length unit
$\mathbf{K}$	Camera intrinsic matrix	3×3; focal entries in pixels
$\mathbf{R}, \mathbf{t}$	World-to-camera rotation and translation	3×3, 3×1
$\mathbf{C}$	Camera center in world coordinates	3×1; $\mathbf{C} = -\mathbf{R}^T\mathbf{t}$
$\mathbf{P}$	Homogeneous camera projection matrix	3×4
$\mathbf{H}, \mathbf{F}, \mathbf{E}$	Homography, fundamental matrix, essential matrix	each 3×3, with different constraints
ℓ, π	Homogeneous line / plane coefficients	3×1 / 4×1

Frames, axes, and transformations

Vectors are columns. Image u increases rightward and v downward. Use right-handed camera axes with positive Z_c forward. Specify any different image-plane convention in the figure and derivation.

Write $\mathbf{T}_{B \leftarrow A}$ for a transform that maps coordinates from frame A into frame B. In a product, the rightmost transform acts first. State whether a rotation acts on points or changes the coordinate frame. Use scalar-first unit quaternions (q_w, q_x, q_y, q_z) and define the rotation convention before applying them.

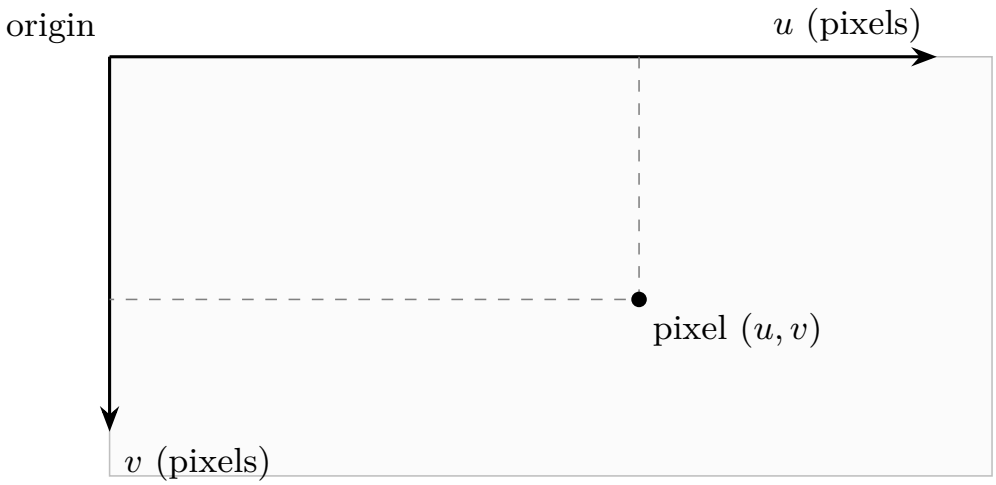


Fig. 1: The pixel-axis convention used throughout the book. The marked pixel is illustrative, not a calibrated measurement. Original diagram; editable TikZ source is retained with the figure assets.

Equality and homogeneous scale

Homogeneous coordinates represent a point by a nonzero vector up to nonzero scale; see *homogeneous coordinates*. Use $=$ for equality of vectors and $\sim$ for equality up to scale. A zero vector is never a projective point.

Formula — homogeneous scale equivalence

$$\mathbf{x} \sim \lambda \mathbf{x}, \qquad \lambda \neq 0. \tag{1}$$

Meaning: scaling all coordinates by the same nonzero number preserves the represented point. Here $\mathbf{x} \in \mathbb{R}^3 \setminus \{\mathbf{0}\}$ and λ is a scalar. This is an exact projective equivalence, not a numerical approximation. When the third coordinate w is nonzero, the corresponding Euclidean point is $(x_1/w, x_2/w)$. When $w = 0$, the point is ideal and this division is invalid.

Application — compare two representations

1. Check that neither vector is zero.
2. Determine whether one nonzero scalar multiplies every component of the first vector to produce the second.
3. For finite points, divide by the last coordinate and compare the resulting Euclidean coordinates.

For example, $(2, 4, 2)^T$ and $(1, 2, 1)^T$ represent the same finite point $(1, 2)$, although the two vectors are not equal. This is an original notation example; lecture-specific derivations and slide references will be added with the chapters.

Numerical checks

Keep pixels distinct from physical lengths. Convert centimeters and meters before substitution. Report exact values before rounded approximations. Distinguish a formula's mathematical conditions from numerical conditioning and state a justified tolerance when checking floating-point results.

You should be comfortable with algebra, trigonometry, basic probability, matrix operations, and introductory calculus. You do not need previous experience with 3D computer vision. Each lecture will refresh the specific tools it uses.

A quick readiness check

- **Vectors:** interpret a dot product, a cross product, and vector length geometrically.
- **Matrices:** check multiplication dimensions, distinguish an inverse from a transpose, and solve a small linear system.
- **Calculus:** interpret a derivative as local sensitivity and a first-order Taylor expansion as a local approximation.
- **Linear algebra:** distinguish eigenvectors from singular vectors and understand a nullspace as the set of solutions to a homogeneous linear system.
- **Probability:** calculate the complement of repeated independent failures.

Attempt the check below. Further numerical refreshers will be added when the corresponding lectures are written.

Check — can these matrices be multiplied?
A matrix A has shape 3×4 and a vector X has shape 4×1. What is the shape of AX? Can XA be formed?

Solution

The inner dimensions of AX agree, so its shape is 3×1. The product XA cannot be formed: its inner dimensions would be 1 and 3. This is the dimension check used later in camera projection.

Part I

Lectures 1–6

CHAPTER

ONE

LECTURE 1: FOUNDATIONS AND 3D GEOMETRY

> **Note**
>
> Planning outline — this lecture has not yet been written. Its full slide audit, examples, and formulas will be completed in lecture order.

1.1 Planned scope

Motivation and applications; 2D/3D geometry; transformations and rotations; camera introduction.

Source deck: `wk1_2_cs532s26_IntroToGeometry.pdf`. Full explanations will record verified printed slide labels and one-based PDF page numbers separately.

1.2 Examples and practice

Select the most useful examples from the actual slides and integrate the assigned questions from the *quiz index*. Missing quiz coverage will be filled with new, clearly labeled worked examples.

1.3 Completion checklist

- Audit every substantive slide topic and formula.
- Write explanations, derivations, and formula/application boxes.
- Add selected slide illustrations and independently checked quiz examples.
- Complete keywords, glossary entries, and source references.
- Check both the website and cumulative printable edition.

CHAPTER

TWO

LECTURE 2: PROJECTIVE GEOMETRY AND TRANSFORMATIONS

> **Note**
>
> Planning outline — this lecture has not yet been written. Its full slide audit, examples, and formulas will be completed in lecture order.

2.1 Planned scope

Geometry review; rotation representations; projectivities and warping; DLT/RANSAC; two-view introduction.

Source deck: `cs532_f25_Week2.pdf`. Full explanations will record verified printed slide labels and one-based PDF page numbers separately.

2.2 Examples and practice

Select the most useful examples from the actual slides and integrate the assigned questions from the *quiz index*. Missing quiz coverage will be filled with new, clearly labeled worked examples.

2.3 Completion checklist

- Audit every substantive slide topic and formula.
- Write explanations, derivations, and formula/application boxes.
- Add selected slide illustrations and independently checked quiz examples.
- Complete keywords, glossary entries, and source references.
- Check both the website and cumulative printable edition.

CHAPTER

THREE

LECTURE 3: CAMERA PROJECTION, HOMOGRAPHIES, AND RANSAC

> **Note**
>
> Planning outline — this lecture has not yet been written. Its full slide audit, examples, and formulas will be completed in lecture order.

3.1 Planned scope

Image formation; camera parameters; homographies; estimation and robust fitting.

Source deck: `wk3_cs532s26_ImageFormation+Homographies-1.pdf`. Full explanations will record verified printed slide labels and one-based PDF page numbers separately.

3.2 Examples and practice

Select the most useful examples from the actual slides and integrate the assigned questions from the *quiz index*. Missing quiz coverage will be filled with new, clearly labeled worked examples.

3.3 Completion checklist

- Audit every substantive slide topic and formula.
- Write explanations, derivations, and formula/application boxes.
- Add selected slide illustrations and independently checked quiz examples.
- Complete keywords, glossary entries, and source references.
- Check both the website and cumulative printable edition.

CHAPTER

FOUR

LECTURE 4: TWO-VIEW GEOMETRY AND STEREO

> **Note**
>
> Planning outline — this lecture has not yet been written. Its full slide audit, examples, and formulas will be completed in lecture order.

4.1 Planned scope

Distortion; epipolar geometry; fundamental and essential matrices; stereo.

Source deck: `wk4_cs532s26_TwoViewGeometry.pdf`. Full explanations will record verified printed slide labels and one-based PDF page numbers separately.

4.2 Examples and practice

Select the most useful examples from the actual slides and integrate the assigned questions from the *quiz index*. Missing quiz coverage will be filled with new, clearly labeled worked examples.

4.3 Completion checklist

- Audit every substantive slide topic and formula.
- Write explanations, derivations, and formula/application boxes.
- Add selected slide illustrations and independently checked quiz examples.
- Complete keywords, glossary entries, and source references.
- Check both the website and cumulative printable edition.

CHAPTER

FIVE

LECTURE 5: CORNERS, TRACKING, AND MATCHING

> **Note**
>
> Planning outline — this lecture has not yet been written. Its full slide audit, examples, and formulas will be completed in lecture order.

5.1 Planned scope

Corner detection; feature tracking; descriptors and matching.

Source deck: `wk5_cs532_Corners.pdf`. Full explanations will record verified printed slide labels and one-based PDF page numbers separately.

5.2 Examples and practice

Select the most useful examples from the actual slides and integrate the assigned questions from the *quiz index*. Missing quiz coverage will be filled with new, clearly labeled worked examples.

5.3 Completion checklist

- Audit every substantive slide topic and formula.
- Write explanations, derivations, and formula/application boxes.
- Add selected slide illustrations and independently checked quiz examples.
- Complete keywords, glossary entries, and source references.
- Check both the website and cumulative printable edition.

CHAPTER

SIX

LECTURE 6: STRUCTURE FROM MOTION

> **Note**
>
> Planning outline — this lecture has not yet been written. Its full slide audit, examples, and formulas will be completed in lecture order.

6.1 Planned scope

Reconstruction ambiguity; factorization; triangulation; bundle adjustment.

Source deck: `wk6_cs532s24_SFM1.pdf`. Full explanations will record verified printed slide labels and one-based PDF page numbers separately.

6.2 Examples and practice

Select the most useful examples from the actual slides and integrate the assigned questions from the *quiz index*. Missing quiz coverage will be filled with new, clearly labeled worked examples.

6.3 Completion checklist

- Audit every substantive slide topic and formula.
- Write explanations, derivations, and formula/application boxes.
- Add selected slide illustrations and independently checked quiz examples.
- Complete keywords, glossary entries, and source references.
- Check both the website and cumulative printable edition.

Study and reference

FORMULA REFERENCE

This reference will grow one lecture at a time. Every entry points to a named formula box, its assumptions, derivation, and application procedure. Lecture formulas have not yet been populated.

Foundation	Formula	Explanation and application
Homogeneous scale	(1)	*Notation and coordinate conventions*

QUIZ PRACTICE INDEX

These are planned placements from `quizByLec.md`. Questions and solutions have not yet been adapted or independently verified. Duplicate sources are retained for later closed-notes retests.

Lecture 1

Fall 2025 Quiz 1

Source: `cs532f25_quiz_1_solution.pdf`

Planned use: Relative poses; plane construction; probability (secondary L3).

Fall 2017 Quiz 9

Source: `cs532_f17_quiz9.pdf`

Planned use: Rotated normals; depth backprojection (extension after L3/4).

Lecture 2

Spring 2020 Quiz 1

Source: `cs532s20_quiz1.pdf`

Planned use: Collinearity; intersections; projection dimensions.

Lecture 3

Spring 2019 Quiz 3

Source: `cs532s19-quiz3.pdf`

Planned use: Pentagon matches; homography; RANSAC; symmetry.

Spring 2019 Quiz 2

Source: `cs532s19_quiz2-2.pdf`

Planned use: FOV; intrinsics; projection; principal point.

Spring 2020 Quiz 3 (mixed printed header)

Source: `cs532s20_quiz3.pdf`

Planned use: Camera calibration retest.

Retest: near-duplicate of cs532s19_quiz2-2.pdf.

Spring 2020 Quiz 2

Source: `cs532s20_quiz2.pdf`

Planned use: Pinhole projection; ray cross product.

Filename 2015 Quiz 9; printed Spring 2019

Source: `cs532f15_quiz9.pdf`

Planned use: Ray-plane depth.

Lecture 4

Filename 2015 Quiz 4; printed Spring 2019

Source: `cs532f15_quiz4.pdf`

Planned use: Fundamental matrix; epipolar constraint.

Spring 2019 Quiz 4

Source: `cs532s19_quiz4.pdf`

Planned use: Epipolar geometry retest.

Retest: duplicate of cs532f15_quiz4.pdf per study notes.

Fall 2025 Quiz 2

Source: `cs532f25_quiz_2_solution.pdf`

Planned use: Three-view lines; epipolar-consistent points; source erratum.

Fall 2025 Quiz 3

Source: `cs532f25_quiz_3_solution.pdf`

Planned use: Stereo depth; error; disparity bounds.

Spring 2019 Quiz 5

Source: `cs532s19_quiz5-2.pdf`

Planned use: Exact positive/negative disparity error.

Spring 2020 Quiz 5

Source: `cs532s20_quiz5.pdf`

Planned use: Disparity error retest.

Retest: duplicate of cs532s19_quiz5-2.pdf per study notes.

Lecture 5

Fall 2025 Quiz 4

Source: `cs532f25_quiz_4_solution.pdf`

Planned use: KLT; exposure normalization.

Spring 2019 Quiz 6

Source: `cs532s19_quiz6-2.pdf`

Planned use: Corners; edges; isolated pixels; aperture problem.

Lecture 6

Fall 2025 Quiz 5

Source: `cs532f25_quiz_5_solution-1.pdf`

Planned use: Reconstruction upgrades; factorization vs sequential.

Fall 2025 Quiz 6

Source: `cs532f25_quiz_6_solution.pdf`

Planned use: Large-scale SfM; contaminated matches; advanced transfer.

Outside the core scope

Fall 2025 Quizzes 7 (ICP) and 8 (meshes) are recorded in the source register and excluded from Lectures 1–6.

GLOSSARY

Terms are added as their explanations are written. First-use definitions, glossary terms, and PDF index entries share stable references.

Homogeneous coordinates

Coordinates representing a projective point by a nonzero vector up to nonzero scale. See *Notation and coordinate conventions* and (1) for the definition and comparison procedure.

Coordinate frame

A reference origin and axes used to express coordinates. Transform directions are specified in *Notation and coordinate conventions*; detailed geometry will follow in Lecture 1.

SOURCES AND FIGURE CREDITS

The planned text follows the six supplied CS 532 lecture PDFs and the quiz-to-lecture mapping in `quizByLec.md`. The lecture decks identify Enrique Dunn and credit additional contributors on individual sections. Preserve those specific credits when adapting their material; do not infer authorship or reuse permission from filenames alone.

Specific lecture citations and image credits will be added when material is selected and adapted. The notation and matrix-dimension examples in this foundation edition are original explanatory examples.

The pixel-axis diagram in the notation guide is an original project illustration with an editable TikZ source and paired SVG/PDF assets.

No third-party lecture images have been imported in this foundation edition. A repository's text license does not automatically establish the reuse permissions of every embedded source figure.

ERRATA AND ASSUMPTIONS TO VERIFY

These source issues are flagged by the supplied study notes. Detailed corrections will be independently checked when the relevant worked examples are written.

Source	Issue	Planned treatment
Fall 2025 Quiz 2	A line-coefficient vector is described as a point	Verify arithmetic, matrix rank, and the scalar epipolar constraint in Lecture 4
Fall 2025 Quiz 5	A planar grid supports a full-scene metric-upgrade claim	State calibration/view assumptions and distinguish plane from full-scene recovery in Lecture 6
`cs532f15_quiz4.pdf`, `cs532f15_quiz9.pdf`	Filenames suggest 2015; printed headers identify Spring 2019	Retain both names in citations
`cs532s20_quiz3.pdf`	Printed header mixes years/quiz labels	Record the filename and printed wording

All source-solution computations remain subject to independent checking. Relevance scores describe slide relevance, not exam probabilities.

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